

Handwritten header for a table: Handwritten X Y (marks)

2	10
3	16
7	37
8	42
6	30
15	75
20	100
12	62
10	50
16	81

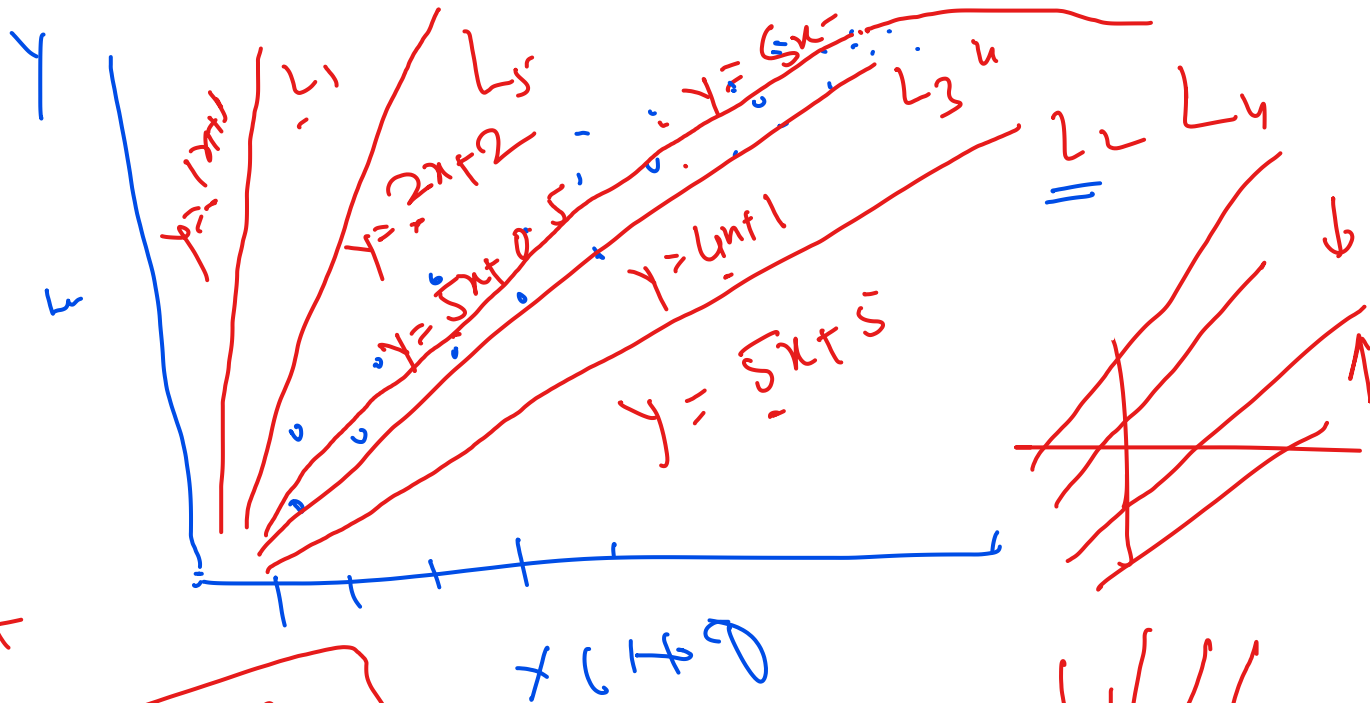
$y = 5x + 5$

$y = 5x + 0.5$
 $75 - 5$

$10 \rightarrow 50$
 $16 \rightarrow 81$

Best

get m, C



Equation of line:-

$y = mx + C$
 slope $\downarrow$ Intercept $\downarrow$

$y = w_1x + b$

15

18

19 =

90 + 0.5
90.5

$L_4 =$

$$\hat{y} = \boxed{5x + 0.5} =$$

$L_3 =$

$$\hat{y} = \underline{4x + 1}$$

- E_1

- E_L
 E_S

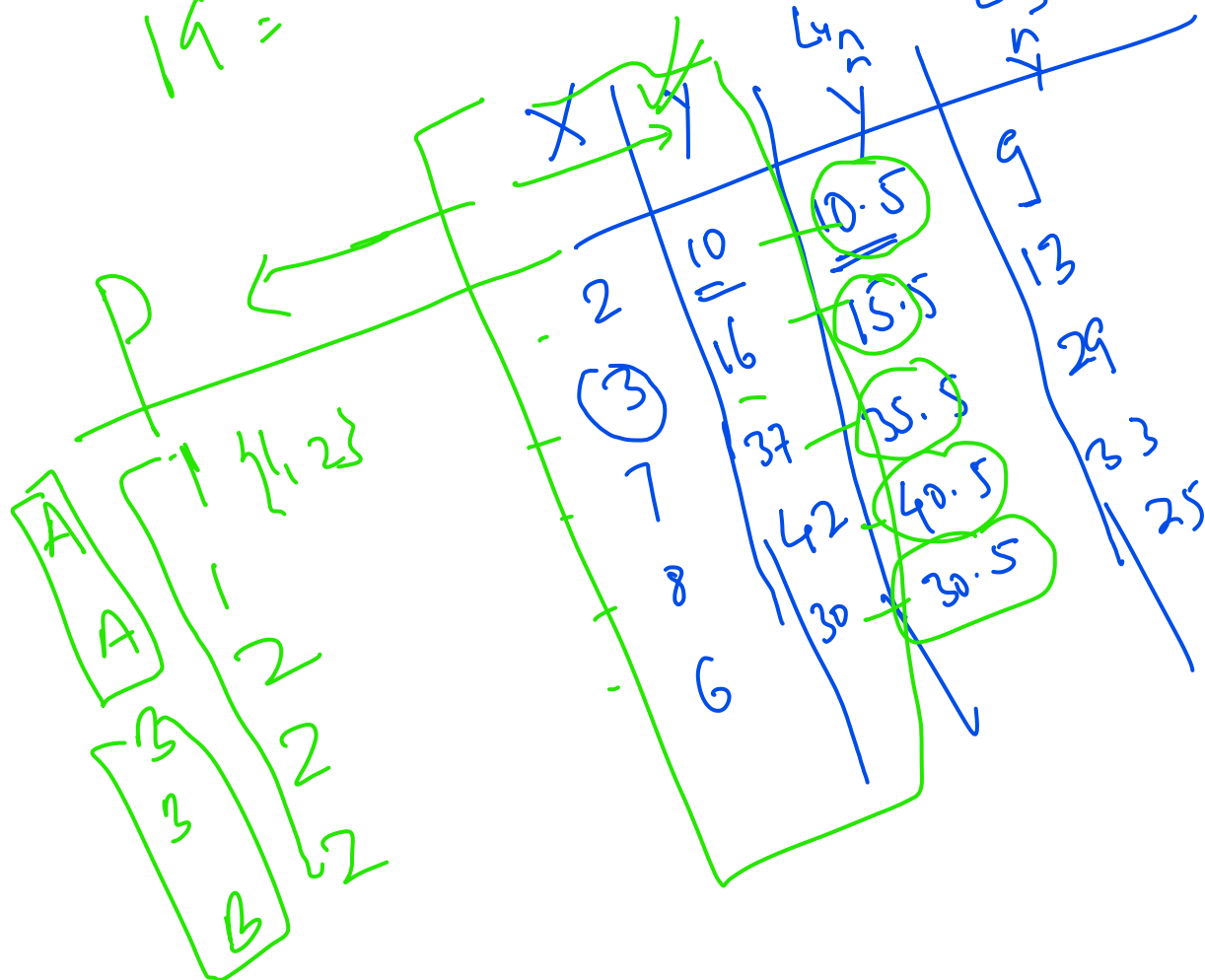


For, $E_4 = \frac{1}{5} \sum (y - \hat{y})^2$

for $E_3 = \frac{1}{5} \sum (y - \hat{y})^2$

My goal = $\boxed{m, C}$

$$y = mx + C$$



X	Y
2	4
3	6
4	8

$$\hat{y} = w_0 + w_1 x$$

Goal is to get
best values
 w_0, w_1

$$J(w) = \frac{1}{n} \sum (y - \hat{y})^2$$

Iteration no 1

$$w_0 = 0$$

$$w_1 = 1$$

$$\hat{y} = 0 + 1 \times x = \boxed{\hat{y} = x}$$

X	Y	$\hat{y}$
2	4	2
3	6	3
4	8	4

Random
Guess method

$$\text{Mean sq error} = \frac{1}{3} \sum (y - \hat{y})^2 = \frac{1}{3}$$

$$= \frac{1}{3} [(4-2)^2 + (6-3)^2 + (8-4)^2] = \frac{29}{3} \text{ (10)}$$

Iteration no 2 $w_0 = 1$ $w_1 = 3$

//

$$\hat{y} = \underline{1 + 3x}$$

x	y	$\hat{y}$
2	4	7
3	6	10
4	8	13

$$\frac{1}{3} [(4-7)^2 + (6-10)^2 + (8-13)^2]$$

$$= \frac{50}{3} = \textcircled{16.2}$$

Iteration no 3

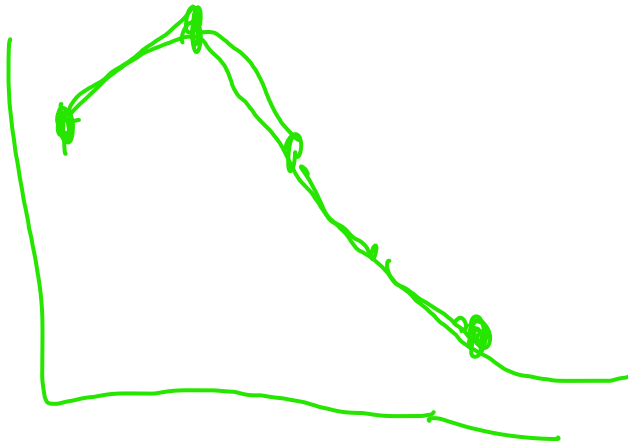
$$\underline{w_0 = 0.5} \quad \underline{w_1 = 2}$$

$$\hat{y} = \underline{0.5 + 2x}$$

x	y	$\hat{y}$
2	4	4.5
3	6	6.5
4	8	8.5

$$\frac{1}{3} [0.25 + 0.25 + 0.25]$$

$$= \textcircled{\boxed{0.25}}$$



Other method w_0, w_1 $\rightarrow$ Gradient descent

Objective function:

$$J(\theta) = \frac{1}{n} \sum (\hat{y} - \tilde{y})^2 \Rightarrow \frac{1}{n} \sum (1 - \theta_0 + \theta_1 x_1 + \theta_2 x_2)^2$$

Target = min the error

$$J(w) = \frac{1}{n} \sum (\hat{y} - \tilde{y})^2 = \frac{1}{n} \sum (1 - w_0 + w_1 x_1 + w_2 x_2 + \dots)^2$$

How

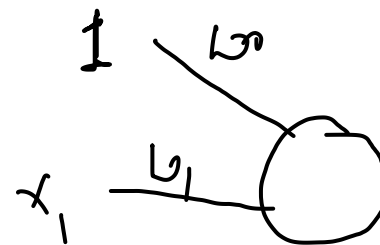
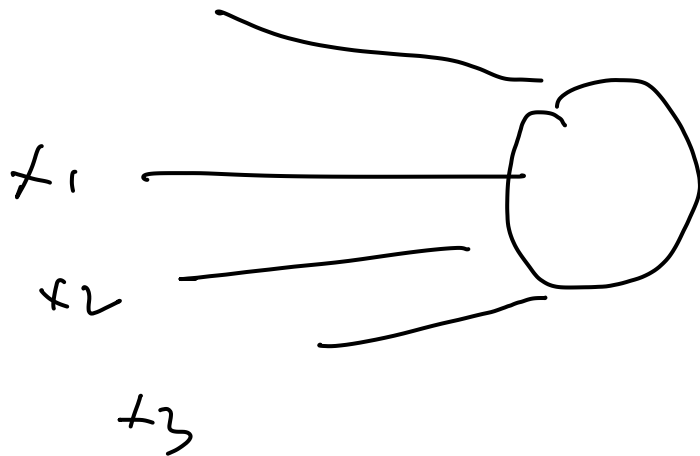
Change values of $(w_0, w_1, w_2, \dots)$

Not using Random Guess using Gradient descent

$$f(\eta) = x+1 \longrightarrow \boxed{2}$$

$$f(\eta) = (x+1)^2 \longrightarrow \boxed{4}$$

$$f(\eta) = \frac{1}{1+e^x} = \frac{1}{1+e^{-100}} = \boxed{0.7}$$



Linear Regression

$$J(w) = \frac{1}{N} \sum \frac{1}{2} (w^T x - y)^2$$

$$J(w) = \frac{1}{N} \sum \frac{1}{2} (w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \dots - y)^2$$

$$\frac{\partial J(w)}{\partial w_1} = \frac{1}{N} \sum \frac{1}{2} x_1^2 (w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \dots - y)$$

(0 + x_1 + 0 + 0 + ...)

$$\frac{\partial J(w)}{\partial w_2} = \frac{1}{N} \sum (w_0 + w_1 x_1 + \dots - y) x_2$$

$$\frac{\partial J(w)}{\partial w_0} = \frac{1}{N} \sum (w_0 + w_1 x_1 + w_2 x_2 \dots - y) (1)$$

$$\frac{\partial J(w)}{\partial w_0} = \frac{1}{N} \sum (w_0 + w_1 x_1 + w_2 x_2 \dots - y)$$

$$w_{old} = \boxed{5}$$

$$\eta = \boxed{0.5}$$

$$\frac{\partial J}{\partial w} = \boxed{10}$$

$$w^{new} = w^{old} - \eta \frac{\partial J}{\partial w}$$

$$w^{new} = 5 - 0.5(10)$$

$$w^{new} = 5 - 5 = \underline{\underline{0}}$$

$$w_{old} = 0$$

$$\eta = \underline{\underline{0.2}}$$

$$\frac{\partial J}{\partial w} = -1$$

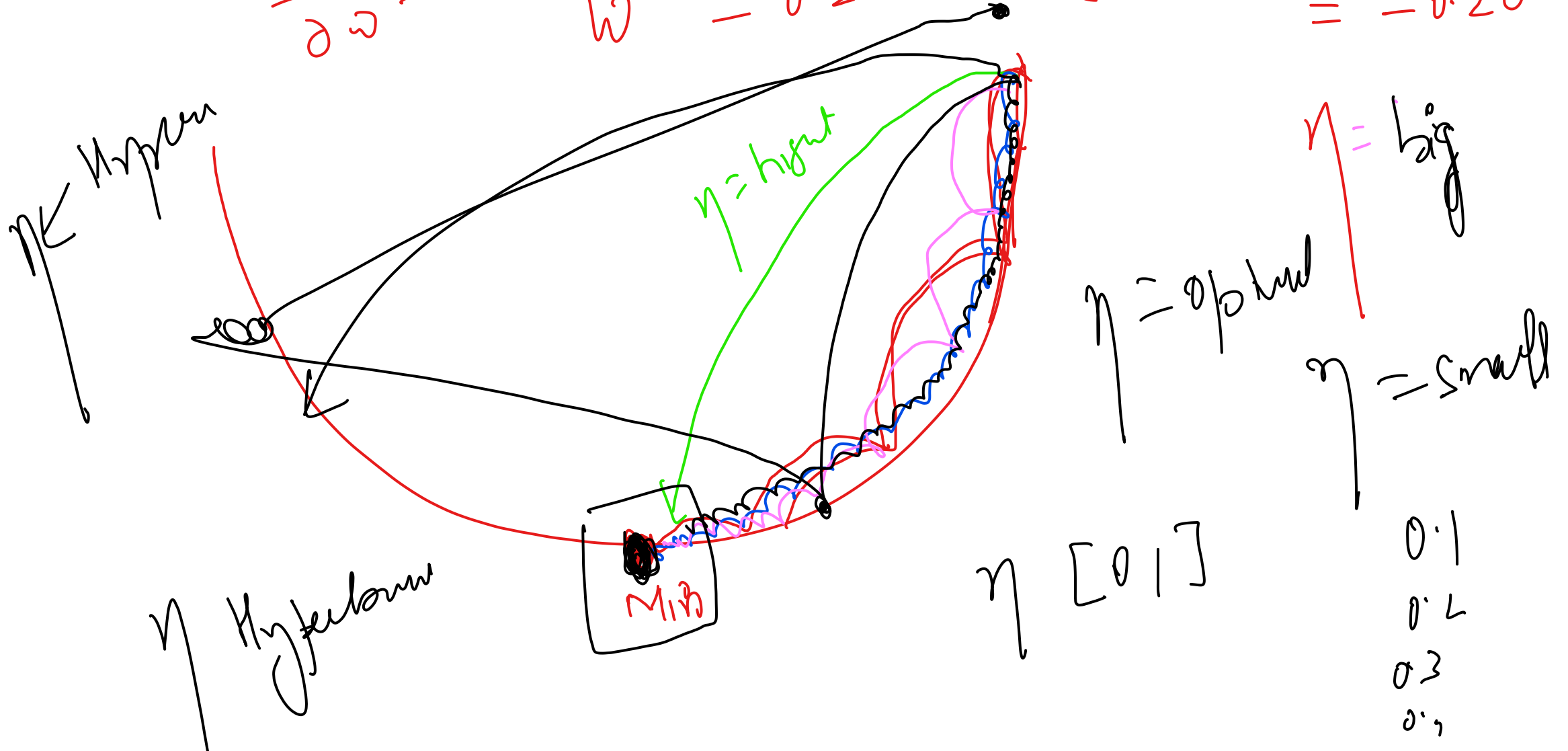
$$w^{new} = w^{old} - \eta \frac{\partial J}{\partial w}$$

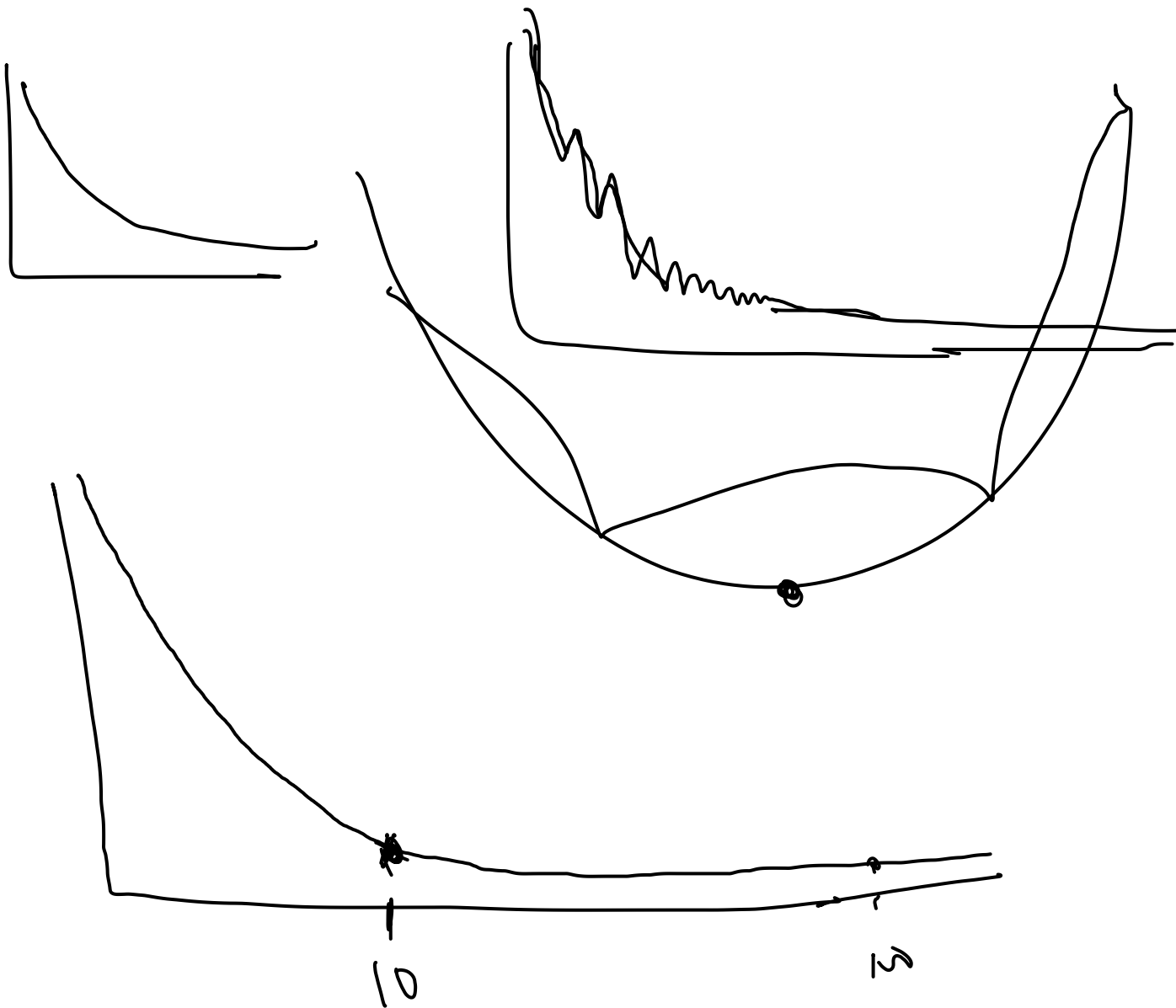
$$w^{new} = 0 - 0.2(-1) = 0.2$$

$$\frac{\partial U}{\partial w} = 0.2$$

$W = 0.2$

$$w^{\text{new}} = 0.2 - 0.2(0.2) = 0.2 - 0.04 = 0.16$$





$\mathcal{E}$

